Baseline-Normalised Engagement Index

Mathematical framing of the AttentivU pipeline and the z-scoring stage

NOVA Buildathon 2026 — COG-BCI / PVT

Scope

This document formalises the algorithmic (non-learned) arm of the attention-lapse pipeline: the AttentivU engagement index [1] computed on COG-BCI PVT trials [2], and in particular the **baseline normalisation** that makes that index comparable across electrodes, sessions and participants.

The learned arm (EEGNet [3]) consumes the same preprocessed tensor and is described in the main research document. The two arms are kept on **identical windows** so that their outputs are directly comparable; §2.1 explains why that constraint is not cosmetic.

Every numeric constant quoted below was measured on the local COG-BCI copy, not taken from the literature. Measurements labelled **sub-01** are single-session and illustrative; measurements labelled with a sample size are aggregates.

Notation

i trial index, $1...N$
c channel index, $1...C$, $C = 62$
w baseline window index, $1...W$, $W \approx 56$
s, e subject and session; $s(i), e(i)$ denote those of trial i
f frequency, Hz
f_s sampling rate after resampling, 128 Hz
T window length in samples, $T = 256$ (2000 ms)
H hop between baseline windows, $H = T/2$

Three indices do three distinct jobs and are never interchangeable: c selects an electrode, w sweeps the baseline windows of one recording, and the pair (s, e) selects which recording. A subject alone does not identify a baseline — each participant sat three sessions and the cap was remounted between them, so there are $25 \times 3 = 75$ distinct baselines and $75 \times 62 = 4650$ calibration pairs.

The pipeline as a whole

Preprocessing operator

Let $x \in \mathbb{R}^{C \times n}$ be a continuous recording. AttentivU steps 1–7 define an operator $\mathcal{P}$ applied **identically** to every recording, task and resting alike:

$$\mathcal{P} = \mathcal{K} \circ \mathcal{B} \circ \mathcal{R} \circ \mathcal{B} \circ \mathcal{N} \tag{1}$$

with $\mathcal{N}$ a 60 Hz notch (10 Hz width), $\mathcal{B}$ a 4–20 Hz zero-phase Butterworth band-pass, $\mathcal{R}$ resampling 500 → 128 Hz, and $\mathcal{K}$ the channel-wise centre–scale–clip $v \mapsto \min(\max((v - \bar{v})/8, -4), 4)$.

Two properties of $\mathcal{P}$ matter downstream. $\mathcal{N}$, $\mathcal{B}$ and $\mathcal{R}$ are linear and time-invariant, so any distortion they introduce is a **fixed multiplicative factor** on band power and cancels in the difference of logarithms taken in §2. $\mathcal{K}$ is not: clipping is a memoryless nonlinearity whose effect depends on the instantaneous amplitude, and it therefore does **not** cancel. §5 quantifies both.

Two sampling domains

The operator output is sampled in two different ways.

Trials. For trial i with stimulus onset t_i and guard $\delta = 100$ ms, extract the sample set ending δ before onset:

$$\mathcal{T}_i = \{t_i - \delta - T + 1, ..., t_i - \delta\} \tag{2}$$

$$X_i = [\mathcal{P}x]_{\cdot,\mathcal{T}_i} \in \mathbb{R}^{C \times T} \tag{3}$$

admitted only if the preceding response cleared by more than $T/f_s + 200$ ms, so that no post-response activity enters the window. This is exactly the tensor the network receives.

Baseline. For the eyes-open pre-task rest run `RS_Beg_EO` of the same session, discard τf_s samples at each end ($\tau = 1$ s) and tile the remainder:

$$a_w = (w - 1)H + \tau f_s, \quad \mathcal{B}_w = \{a_w + 1, ..., a_w + T\} \tag{4}$$

$$R_{s,e,w} = [\mathcal{P}r]_{\cdot,\mathcal{B}_w} \tag{5}$$

for $w = 1...W$ with $W = \lfloor (n - 2\tau f_s - T)/H \rfloor + 1$, so the baseline windows have exactly the length and sample count of a trial window.

The trim is not arbitrary: the impulse response of the zero-phase Butterworth cascade decays below 10^{-3} of peak within 0.88 s at 128 Hz, so the terminal windows of a continuous run are part filter transient. Trials are exempt — the first PVT stimulus occurs at ≈ 10.9 s.

Spectral operator

A single estimator $\mathcal{W}$ is applied to both domains: Welch [4] with Hamming taper, segment length $L = f_s$ (1 s), 50% overlap, zero-padded to $N_{\text{fft}} = T$, averaged over the $K = 3$ resulting segments, DC removed, evaluated on $[4, 20]$ Hz at $\Delta f = f_s / N_{\text{fft}} = 0.5$ Hz.

Fixing L **independently of** T decouples the estimator from window length. Averaging $K = 3$ segments reduces estimator variance: measured window-to-window spread of $\ln E$ falls from 0.428 ($K = 1$, a single periodogram) to 0.309.

Engagement functional

Band powers by trapezoidal integration over **half-open** intervals:

$$P_b = \int_{[f_1, f_2)} \hat{P}(f) \, \mathrm{d}f \tag{6}$$

for $\theta = [4, 7)$, $\alpha = [7, 11)$, $\beta = [11, 20)$ Hz. Half-open matters: closed intervals on both edges assign 7 Hz to both θ and α and 11 Hz to both α and β, inflating θ by 21% on a realistic $1/f$-plus-alpha spectrum.

The index of Pope et al. [5], in log space:

$$\ell = \ln P_\beta - \ln(P_\alpha + P_\theta) \tag{7}$$

Two remarks. ℓ is **scale-invariant**: multiplying a channel by any constant leaves it unchanged, so the $\div 8$ inside $\mathcal{K}$ has no effect whatsoever — only the clip does. And ℓ rather than E is the quantity to standardise: E is a ratio of near-log-normal powers, hence right-skewed with dispersion dominated by the upper tail, whereas ℓ is near-symmetric and admits a location–scale description.

Applying $\mathcal{W}$ then Equation 7 gives $\ell_{i,c}$ for trials and $\ell^{\text{rest}}_{s,e,w,c}$ for baseline windows.

The reduction, dimension by dimension

Each ℓ is an entire window collapsed to one number. The time axis is destroyed at the spectral step and never reappears:

```
rest run        (62, 7674)     ch x samples
  | tile
windows         (56, 62, 256)  +window axis
  | Welch
spectra         (56, 62, 33)   time -> freq
  | integrate,log
log engagement  (56, 62)       1 per win-ch
  | fix channel c
x               (56,)          the median set
```

Concretely, for $w = 0$, channel Pz of sub-01/ses-S1: the 256 voltages give $P_\theta = 0.2214$, $P_\alpha = 0.2969$, $P_\beta = 0.6723$, hence $E = 1.2970$ and $\ell = 0.2600$. That single scalar is one element of the 56-element set standardised below.

The z-score

Why standardise at all

E has no absolute meaning. Band power scales with skull thickness, hair, electrode impedance and cap placement, so the same numeric value at two electrodes, or on two days, does not denote the same neural state. The pipeline therefore never asks **what is the value** but **how far is this value from what this electrode reads, in this session, at rest** — which requires both a location and a scale estimated from a reference distribution.

Stratification

Location and scale are estimated **per (s, e, c)**: per subject, per session, per channel.

Per channel, because resting $P_\beta / (P_\alpha + P_\theta)$ is genuinely site-dependent. Per session, because impedance and placement do not persist across days. Not pooled across subjects, because between-subject variance is precisely the nuisance the baseline exists to remove; pooling σ would leave it in the numerator and let a classifier learn subject identity instead of state. `variance_decomposition()` reports the ICC that quantifies this.

Location and scale

For fixed (s, e, c), write $x_w = \ell^{\text{rest}}_{s,e,w,c}$, $w = 1 \ldots W$. Then

$$\mu_{s,e,c} = \text{median}_w \, x_w \tag{8}$$

$$\sigma_{s,e,c} = 1.4826 \cdot \text{median}_w |x_w - \text{median}_w \, x_w| \tag{9}$$

with the guard $\sigma \leftarrow \max(\sigma, 10^{-3} \, \text{median}_c \, \sigma)$ against degenerate channels. Note the median in Equation 9 is nested: the inner one supplies the centre, the outer one summarises the distances from it. That structure is not exotic — it is the structure of a standard deviation with two substitutions:

$$s = \sqrt{\text{mean}_w \, (x_w - \text{mean} \, x)^2}$$

$$\text{MAD} = \text{median}_w |x_w - \text{median} \, x|$$

Both say: pick a centre, measure each point's distance from it, summarise those distances. The standard deviation uses the mean in both roles and squares; the MAD uses the median in both roles and takes absolute values.

The constant is $1.4826 = 1/\Phi^{-1}(0.75)$, which renders the MAD Fisher-consistent with σ under normality [6]. Without it, every z would be inflated by that factor.

Standardisation

$$z_{i,c} = \frac{\ell_{i,c} - \mu_{s(i),e(i),c}}{\sigma_{s(i),e(i),c}} \tag{10}$$

Only μ and σ survive the baseline; the W windows are discarded. $z_{i,c} = 0$ means the trial's engagement at that electrode matched the session's resting median; $z = -1$ means one resting MAD-σ below it.

Worked example

Channel Pz, sub-01/ses-S1. The $W = 56$ values of ℓ^{rest}:

```
 0.260  0.942  0.498  0.414 -0.020 -0.225  0.625  0.053
-0.652 -1.004 -0.843  0.481  0.041  0.370  0.178 -0.551
-1.514 -1.683 -0.603 -0.534 -0.503 -0.816 -0.887 -0.813
-0.228  0.141  0.155 -0.267  0.077  0.745  0.032 -0.278
-0.404 -1.096 -1.411 -1.624 -1.878 -1.278 -0.645 -1.290
-1.193 -0.782 -1.175 -1.304 -1.609 -0.610  0.414 -0.662
-0.641 -0.566  0.019 -0.727 -0.729 -0.066  0.200 -0.331
```

(A) Sort; with W even the median is the mean of entries 27 and 28: $\mu = (-0.5514 + -0.5341)/2 = -0.5427$. The arithmetic mean would give -0.4606, pulled by the low cluster in windows 32–47.

(B, C) Deviations $x_w - \mu$, then absolute values: $0.803, 1.484, 1.041, 0.957, 0.523, 0.318, \ldots$

(D) Median of those distances: $\text{MAD} = (0.5616 + 0.5745)/2 = 0.5681$. Half the resting windows lie within 0.568 of the centre.

(E) $\sigma = 1.4826 \times 0.5681 = 0.8422$. Empirically 75% of baseline windows fall inside $\pm 1\sigma$ (Gaussian: 68%).

(F) Any trial can now be scored:

trial ℓ	arithmetic	z
-0.543	$(-0.543 + 0.543)/0.842$	0.00
-1.385	$(-1.385 + 0.543)/0.842$	-1.00
-1.600	$(-1.600 + 0.543)/0.842$	-1.26
$+0.300$	$(+0.300 + 0.543)/0.842$	$+1.00$

Why median and MAD rather than mean and standard deviation

Corrupting a single one of the 56 windows with a blink-sized artefact (+12 log units) and refitting:

	centre	1.4826 MAD	s
clean	-0.5427	0.8422	0.6821
one bad window	-0.5427	0.8422	1.8306
change	0.0%	0.0%	+168%

The robust pair is unmoved — a single outlier cannot change which value sits in the middle. The standard deviation nearly triples, and since every trial is divided by it, that one artefact would compress all 90 of this channel's trial z-scores toward zero by a factor of 2.7.

Caveat. The constant 1.4826 assumes approximate normality. On this channel the resting distribution is platykurtic (windows 32–47 sit systematically lower), so $1.4826\,\text{MAD} = 0.842$ exceeds $s = 0.682$ by 23%, where on the channel-median they agree (0.556 vs 0.555). Neither is "the true σ"; the argument for the robust pair is the table above, not superior accuracy.

Channel aggregation

Aggregation happens **after** standardisation, never before: the μ_c differ systematically by site, so averaging raw ℓ across channels would combine incommensurable offsets.

The channel mean of C unit-dispersion variates is not itself unit-dispersion unless the channels are independent. Its scale is therefore estimated on the same baseline windows:

$$g_{s,e,w} = \frac{1}{C} \sum_c \frac{\ell^{\text{rest}}_{s,e,w,c} - \mu_{s,e,c}}{\sigma_{s,e,c}} \tag{11}$$

$$\sigma^G_{s,e} = 1.4826 \cdot \text{MAD}_w\, g_{s,e,w} \tag{12}$$

giving one score per trial,

$$\bar{Z}_i = \frac{1}{\sigma^G_{s(i),e(i)}} \cdot \frac{1}{C} \sum_{c=1}^{C} z_{i,c} \tag{13}$$

Measured $\sigma^G \in [0.69, 1.00]$ across four sessions, against $C^{-1/2} = 0.127$ under independence, because the mean inter-channel correlation of z is 0.47. The effective dimensionality is

$$n_{\text{eff}} = \left(\sigma^G\right)^{-2} \approx 1 \text{ to } 2 \tag{14}$$

Consequence for weighting. With 1–2 effective dimensions there is essentially nothing for a weighting scheme to exploit, and the schemes agree accordingly (correlation across 90 trials): uniform mean vs median $r = 0.988$, vs 20% trimmed mean $r = 0.998$, vs the Pope electrode subset $r = 0.818$. The uniform mean is therefore used. Fitting weights is deliberately avoided here — a parameter-free algorithmic arm is what makes a later comparison against attention pooling interpretable.

Validation

Three checks, in order of precedence.

1. **Construct validity.** $\bar{Z}$ of `RS_Beg_EC` scored against the `RS_Beg_EO` baseline. Eyes-closed alpha enters the denominator of Equation 7, so this must be clearly negative. Measured sub-01: -0.84σ, against -0.01σ for held-out eyes-open windows. If this fails, nothing downstream is interpretable.
2. **Stratification necessity.** ICC of raw ℓ across subjects, from `variance_decomposition()`. A high value confirms that pooling σ across subjects would have been inadmissible.
3. **Primary contrast.** $\mathbb{E}\left[\bar{Z} \mid y = 1\right] - \mathbb{E}\left[\bar{Z} \mid y = 0\right]$, paired within session and aggregated across sessions, where $y = 1$ marks the within-session slowest decile of reaction time.

Assumption audit

Band-edge attenuation. The 4–20 Hz band-pass is applied twice with zero phase, giving measured power transfer of −11.6 dB at 4 Hz and −11.8 dB at 20 Hz — roughly 7% of power surviving at both ends of the analysis range. Because this is LTI and identical on both sides of Equation 10, it cancels in the difference of logarithms to first order; the residual effect on E is $\times 1.07$. The cost is signal-to-noise, not bias.

Window-length coupling. E is strongly dependent on segmentation: the same resting signal yields $\bar{E} = 0.4219$ measured in 2 s windows and 0.2397 in 5 s windows, a 43% shift from segmentation alone. This is why the baseline is tiled at exactly T and not at the 5 s of the original AttentivU formulation. Restoring 5 s windows would additionally require a 5200 ms clear gap, retaining only 68% of trials (61.3 vs 89.8 per session, $n = 29$).

Clipping. $\mathcal{K}$ is nonlinear and amplitude-dependent: the correlation between window peak amplitude and induced $\Delta\ell$ is $+0.44$, so this bias does **not** cancel. It engages on 62% of PVT trials and 10.7% of (trial, channel) pairs, concentrated occipitally where alpha is largest. Mean effect $+0.009$ in high-alpha windows versus 0.000 in low-alpha; worst case 0.25σ. This is the one preprocessing step whose bias reaches the z-score.

Baseline non-stationarity. Channels that degrade **between** the rest block and the PVT shift $\bar{Z}$ substantially (-0.75σ at 12 affected channels, -1.47σ at 24), but the correlation with the uncontaminated composite remains ≥ 0.995: the effect is a near-constant offset, which cancels in a within-session contrast. Channels that are simply bad **throughout** are absorbed almost exactly ($+0.009\sigma$ at 24 affected channels), because the baseline sees the same corruption the trials do.

Implementation map

All of §2–§4 lives in `scripts/dataproc/prototype/engagement.py`.

`welch`	$\mathcal{W}$, §1.3 — shared by both domains
`band_power`	P_b, half-open integration
`engagement_index`	Equation 7
`epoch`	baseline tiling + edge trim, §1.2
`baseline_from_rest`	Equation 8, Equation 9, σ^G
`fit_baselines`	one `Baseline` per (s, e)
`normalize`	Equation 10, returns $N \times 62$
`normalize_composite`	Equation 13, returns N
`baseline_sanity_check`	validation 1
`variance_decomposition`	validation 2
`lapse_contrast`	validation 3

Window length and hop are derived from `config.SAMPLE_SIZE` rather than restated, so the baseline cannot silently desynchronise from the trial tensor that `build_pvt.py` produces. `build_rest.py` must be run with `DATASET = "RS_Beg_EO"` for the baseline and `"RS_Beg_EC"` for validation 1.

References

[1] N. Kosmyna and P. Maes, "AttentivU: An EEG-Based Closed-Loop Biofeedback System for Real-Time Monitoring and Improvement of Engagement for Personalized Learning," *Sensors*, vol. 19, no. 23, 2019, doi: 10.3390/s19235200.

[2] M. F. Hinss, E. S. Jahanpour, B. Somon, L. Pluchon, F. Dehais, and R. N. Roy, *COG-BCI database: A multi-session and multi-task EEG cognitive dataset for passive brain-computer interfaces.* (Jul. 2022). Zenodo. doi: 10.5281/zenodo.6874129.

[3] V. J. Lawhern, A. J. Solon, N. R. Waytowich, S. M. Gordon, C. P. Hung, and B. J. Lance, "EEGNet: a compact convolutional neural network for EEG-based brain–computer interfaces," *Journal of Neural Engineering*, vol. 15, no. 5, p. 56013, Jul. 2018, doi: 10.1088/1741-2552/aace8c.

[4] P. D. Welch, “The use of fast Fourier transform for the estimation of power spectra: A method based on time averaging over short, modified periodograms,” *IEEE Transactions on Audio and Electroacoustics*, vol. 15, no. 2, pp. 70–73, 1967, doi: 10.1109/TAU.1967.1161901.

[5] A. T. Pope, E. H. Bogart, and D. S. Bartolome, “Biocybernetic system evaluates indices of operator engagement in automated task,” *Biological Psychology*, vol. 40, no. 1–2, pp. 187–195, 1995, doi: 10.1016/0301-0511(95)05116-3.

[6] P. J. Rousseeuw and C. Croux, “Alternatives to the Median Absolute Deviation,” *Journal of the American Statistical Association*, vol. 88, no. 424, pp. 1273–1283, 1993, doi: 10.1080/01621459.1993.10476408.